

Experiment 1 - Clock and Periodic Signal Generation

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Abstract— This report explores the LTspice simulation of three timing circuits: a CMOS ring oscillator for measuring inverter propagation delay, an LM555 astable multivibrator for analyzing frequency and duty cycle, and an RC Schmitt trigger oscillator for evaluating its time constant multiplier (α). The results successfully validate theoretical timing equations.

Keywords—Ring Osillator, LM555 Timer, Propagation Delay, Duty Cycle, LTspice Simulation

I. INTRODUCTION

This experiment investigates fundamental oscillator and timing circuits using LTspice, bridging theoretical calculations with practical transient behavior. The study comprises three sections: evaluating inverter propagation delay via a CMOS ring oscillator, analyzing the impact of resistor variations on an LM555 timer’s frequency and duty cycle, and examining the relationship between RC constants and output frequency in a Schmitt trigger oscillator. Ultimately, it provides a comprehensive understanding of timing mechanisms in digital electronics.

II. RING OSCILLATOR

A. Waveform and Frequency

Report the observed waveform along with the measured frequency of the chain.

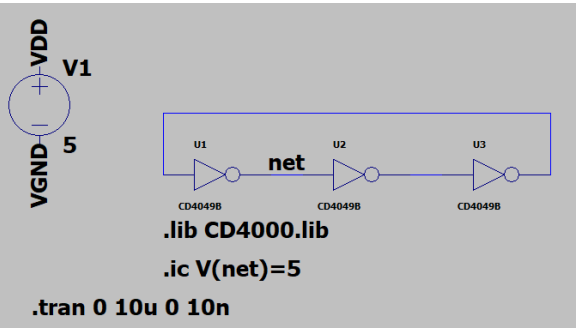


Fig. 1 Schematic of circuit

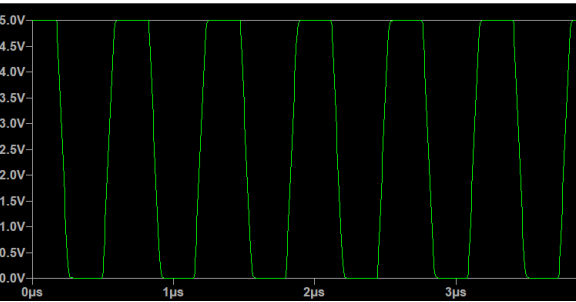


Fig. 2 Waveform

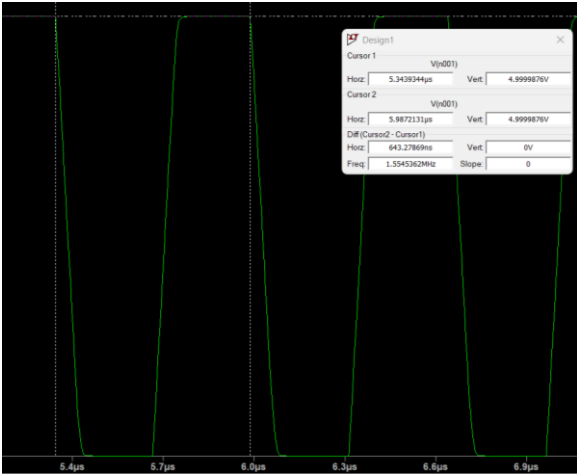


Fig. 3 Measure frequency

$$T \approx 643n, f \approx 1.55MHz$$

$$f = \frac{1}{T} = \frac{1}{0.6 \times 10^{-6}} \approx 1.55 \text{ MHz}$$

B. Delay of a Single Inverter

From previous part we measure the propagation delay of the chain by measuring the period time of the output, so time of a complete period(1 Nt for invert and another one for back to first value = two half cycle) would be:

$$T = 2 \times N \times t_d \rightarrow t_d = \frac{T}{2 \cdot N} \rightarrow t_d = \frac{643ns}{2 \times 3} \approx 107ns$$

C. Compare Delay

Compare the delay of a single inverter with the delay in 7404 TTL specification.

Switching Characteristics

at $V_{CC} = 5V$ and $T_A = 25^\circ C$

Symbol	Parameter	Conditions	Min	Max	Units
t_{PLH}	Propagation Delay Time LOW-to-HIGH Level Output	$C_L = 15 \text{ pF}$ $R_L = 400\Omega$		22	ns
t_{PLH}	Propagation Delay Time HIGH-to-LOW Level Output			15	ns

Fig. 4 table of 7404TTL specification delay from 471s04 pdf

The gate delay time of the CD409B which was found to be 107ns is much longer than the gate transition time of the 7404TTL which is about 20ns. Therefore, the 7404TTL have a much higher switching speed compared to the CD409B (at the same voltage).

D. Upload File

Design1.asc 5/4/2026 4:53 PM ASC File 1 KB

Fig. 5 Design1.asc schematic file from LTspice

III. LM555 TIMER

A. Simulate with Different Resistors

1) $1k\Omega$:

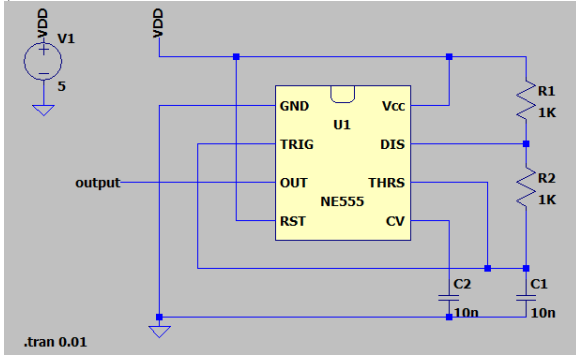


Fig. 6 Schematic of LM555 circuit with 1k resistor

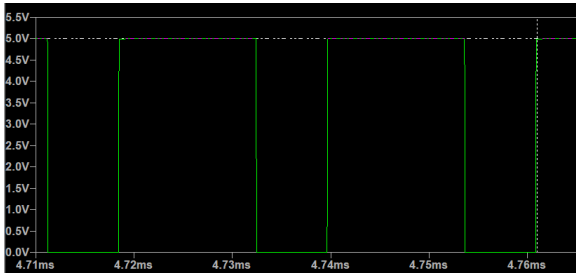


Fig. 7 Waveform of LM555 circuit with 1k resistor

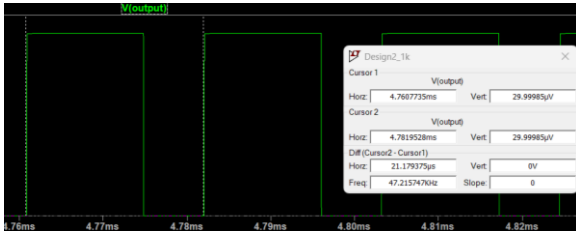


Fig. 8 Measure frequency of LM555 circuit with 1k resistor

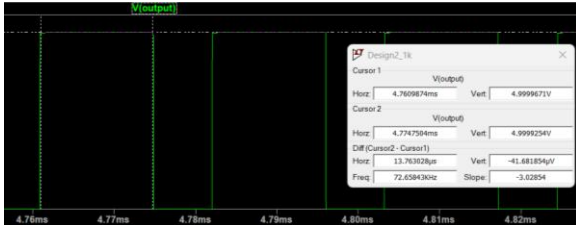


Fig. 9 Measure t_{rise} of LM555 circuit with 1k resistor

$$T \approx 21.18\mu s, f \approx 47.22kHz, t_{rise} \approx 13.76\mu s$$

$$duty\ cycle = \frac{t_{rise}}{T} \times 100 = \frac{13.76\mu s}{21.18\mu s} \times 100 \approx 64.97\%$$

Calculate the expected frequency and duty cycle to verify measurement.

$$f = \frac{1}{T} = \frac{1}{0.693 \times (R_1 + 2R_2) \times C} \rightarrow$$

$$f = \frac{1}{0.693 \times (1k + 2k) \times 10nF} \approx 48.1\ kHz$$

$$duty\ cycle = \frac{R_1 + R_2}{R_1 + 2R_2} = \frac{1 + 1}{1 + 2} \approx 66.67\%$$

2) $10k\Omega$:

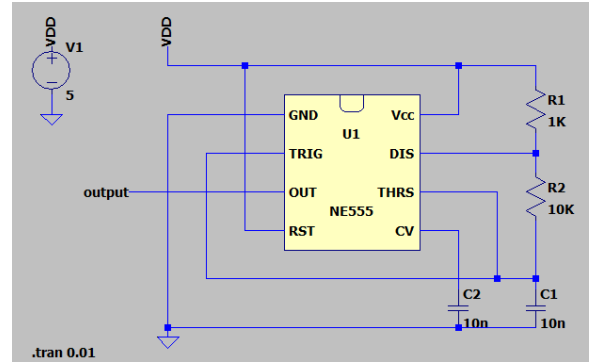


Fig. 10 Schematic of LM555 circuit with 10k resistor

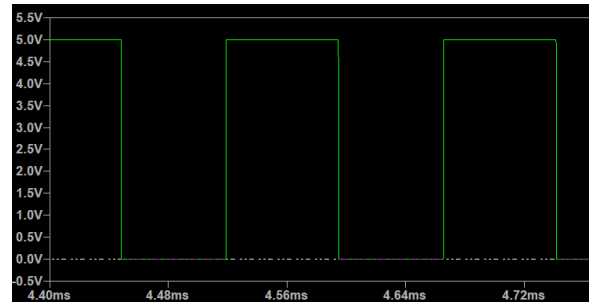


Fig. 11 Waveform of LM555 circuit with 10k resistor

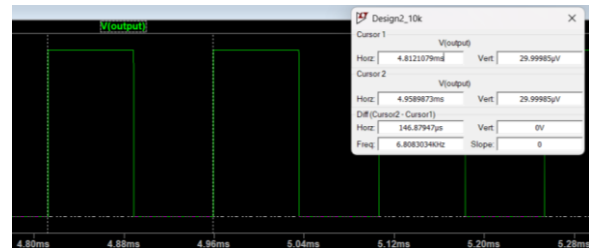


Fig. 12 Measure frequency of LM555 circuit with 10k resistor

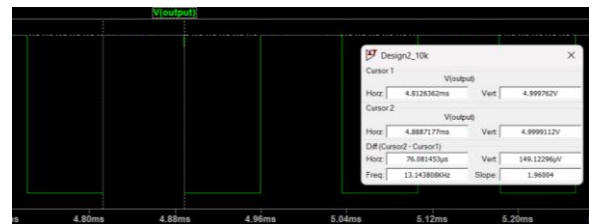


Fig. 13 Measure t_{rise} of LM555 circuit with 10k resistor

$$T \approx 146.87\mu s, f \approx 6.8kHz, t_{rise} \approx 76.08\mu s$$

$$duty\ cycle = \frac{t_{rise}}{T} \times 100 = \frac{76.08\mu s}{146.87\mu s} \times 100 \approx 51.8\%$$

Calculate the expected frequency and duty cycle to verify your measurement.

$$f = \frac{1}{T} = \frac{1}{0.693 \times (R_1 + 2R_2) \times C} \rightarrow$$

$$f = \frac{1}{0.693 \times (1k + 20k) \times 10nF} \approx 6.87 \text{ kHz}$$

$$\text{duty cycle} = \frac{R_1 + R_2}{R_1 + 2R_2} = \frac{1 + 10}{1 + 20} \approx 52.38\%$$

3) 100kΩ:

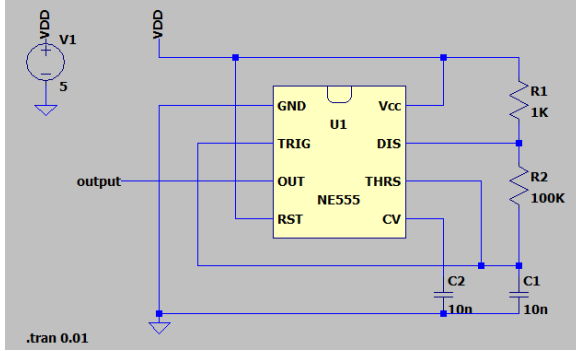


Fig. 14 Schematic of LM555 circuit with 100k resistor

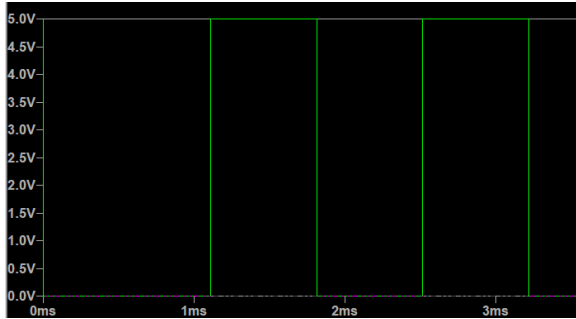


Fig. 15 Waveform of LM555 circuit with 100k resistor

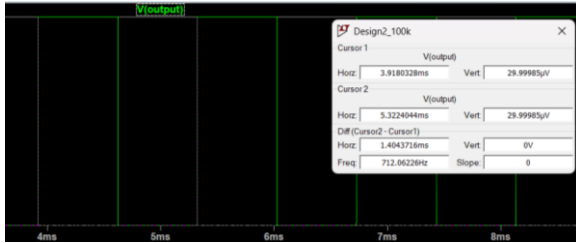


Fig. 16 Measure frequency of LM555 circuit with 100k resistor

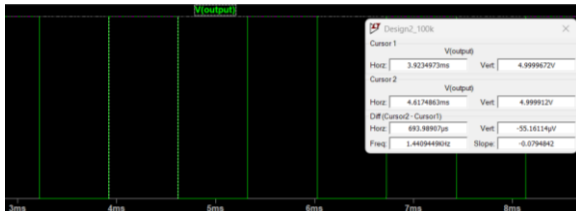


Fig. 17 Measure t_{rise} of LM555 circuit with 100k resistor

$$T \approx 1.4ms, \quad f \approx 712.05Hz, \quad t_{rise} \approx 693.98\mu s$$

$$\text{duty cycle} = \frac{t_{rise}}{T} \times 100 = \frac{693.98\mu s}{1.4ms} \times 100 \approx 49.57\%$$

Calculate the expected frequency and duty cycle to verify your measurement.

$$f = \frac{1}{T} = \frac{1}{0.693 \times (R_1 + 2R_2) \times C} \rightarrow$$

$$f = \frac{1}{0.693 \times (1k + 200k) \times 10nF} \approx 717.91Hz$$

$$\text{duty cycle} = \frac{R_1 + R_2}{R_1 + 2R_2} = \frac{1 + 100}{1 + 200} = 50.25\%$$

B. Effect of Resistance Change on Duty Cycle

According to the relationship $\text{duty cycle} = \frac{R_1 + R_2}{R_1 + 2R_2}$, since the resistance R_1 is constant (1 kΩ), with a significant increase in the value of the resistance R_2 , the term R_1 in the numerator and denominator of the fraction against R_2 becomes negligible. As a result, the fraction tends to $\frac{R_2}{2R_2} = \frac{1}{2}$. Therefore, the more $R_2 \gg R_1$, the duty cycle gets closer to 50% (as we saw in the calculations and simulations, it went from 64.97% to 51.8%).

C. Upload Files

Design2_1k.asc	5/4/2026 10:06 PM	ASC File	2 KB
Design2_10k.asc	5/4/2026 10:14 PM	ASC File	2 KB
Design2_100k.asc	5/4/2026 10:34 PM	ASC File	2 KB

Fig. 18 LTspice schematic files for variation of R_2 resistors: *Design2_1k.asc*, *Design2_10k.asc* and *Design2_100k.asc*

IV. SCHMITT TRIGGER OSCILLATOR

A. Frequency and a Parameter for Different Resistors

1) 470 Ω:

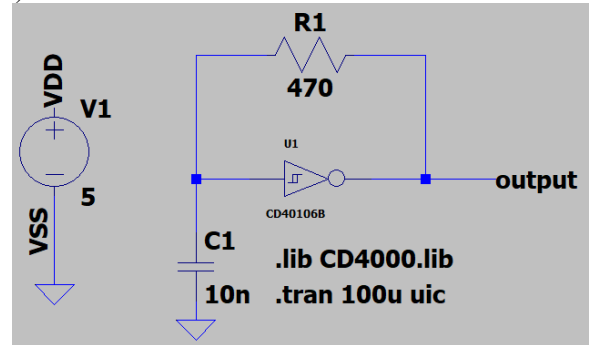


Fig. 19 Schematic of STO circuit with 470 resistor

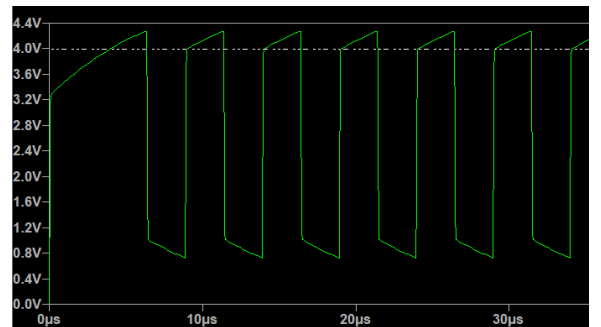


Fig. 20 Waveform of STO circuit with 470 resistor

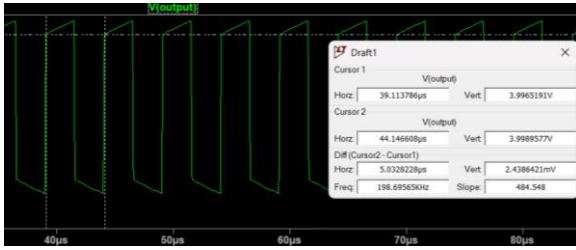


Fig. 21 Measure frequency of STO circuit with 470 resistor

$$f = \frac{\alpha}{RC} \rightarrow \alpha = fRC \rightarrow \alpha = 198.69k * 470 * 10n \approx 0.93$$

2) 1 kΩ:

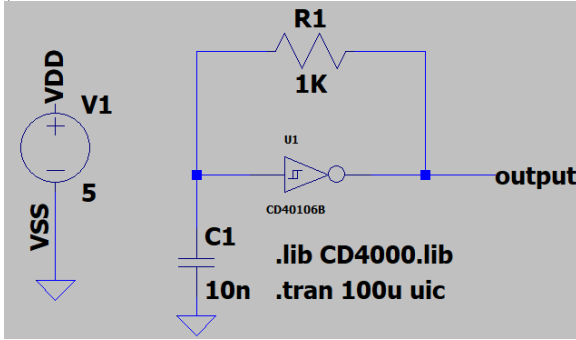


Fig. 22 Schematic of STO circuit with 1k resistor

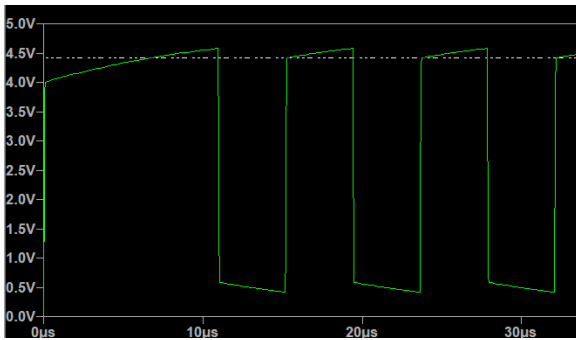


Fig. 23 Waveform of STO circuit with 1k resistor

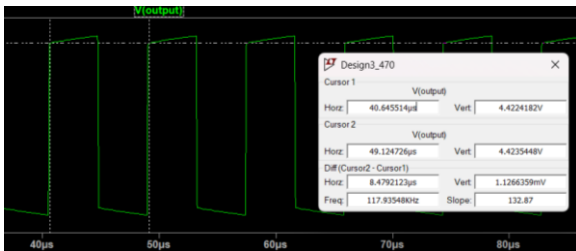


Fig. 24 Measure frequency of STO circuit with 1k resistor

$$f = \frac{\alpha}{RC} \rightarrow \alpha = fRC \rightarrow \alpha = 117.94k * 1k * 10n \approx 1.17$$

3) 2.2 kΩ:

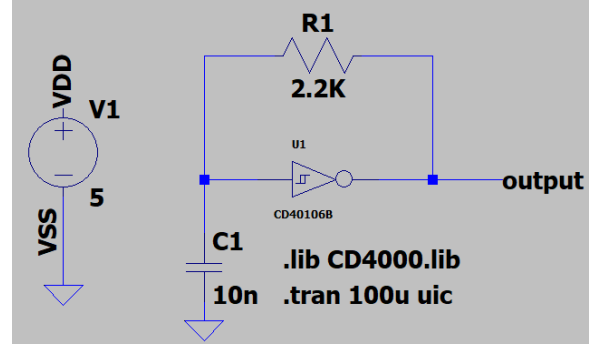


Fig. 25 Schematic of STO circuit with 2.2k resistor

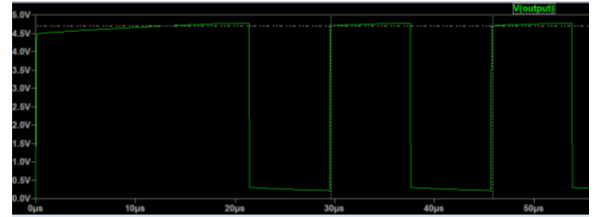


Fig. 26 Waveform of STO circuit with 2.2k resistor

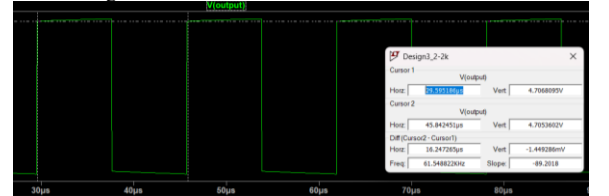


Fig. 27 Measure frequency of STO circuit with 2.2k resistor

$$f = \frac{\alpha}{RC} \rightarrow \alpha = fRC \rightarrow \alpha = 61.55k * 2.2k * 10n \approx 1.35$$

B. Report the Average α

$$\alpha = \frac{\alpha_1 + \alpha_2 + \alpha_3}{3} = \frac{0.93 + 1.17 + 1.35}{3} = \frac{3.45}{3} = 1.15$$

C. Upload the LTspice Schematic Files

Design3_1k.asc	5/4/2026 11:37 PM	ASC File	1 KB
Design3_2-2k.asc	5/4/2026 11:39 PM	ASC File	1 KB
Design3_470.asc	5/4/2026 11:34 PM	ASC File	1 KB

Fig. 28 LTspice schematic files for variation of R resistors: Design3_470.asc, Design3_1k.asc and Design3_2.2k.asc